

MOXIFLOXACIN OPHTHALMIC SOLUTION 0.5% w/v MICROMOX EYE DROPS

BRAND OR PRODUCT NAME
MICROMOX EYE DROPS

NAME AND STRENGTH OF ACTIVE SUBSTANCE(S)
Moxifloxacin 0.5% w/v

PRODUCT DESCRIPTION
Clear, greenish- yellow solution practically free from particles

PHARMACODYNAMICS:

Pharmacotherapeutic group: Ophthalmologicals; anti-infective, other anti-infective, ATC code: S01A E07

Mode of Action:

Moxifloxacin, a fourth-generation fluoroquinolones, inhibits the DNA gyrase and topoisomerase IV required for bacterial DNA replication, repair, and recombination.

Resistance:

Resistance to fluoroquinolones, including moxifloxacin generally occurs by chromosomal mutations in genes encoding DNA gyrase and topoisomerase IV. In Gram-negative bacteria, moxifloxacin resistance can be due to mutations in *mar* (multiple antibiotic resistances) and the *qnr* (quinolone resistance) gene systems. Resistance is also associated with expression of bacteria efflux proteins and inactivating enzymes. Cross-resistance with beta-lactams, macrolides and aminoglycosides is not expected due to differences in mode of action.

Susceptibility Testing Breakpoints

There are no pharmacological data correlated with clinical outcome for moxifloxacin administered as a topical agent. As a result, the European Committee on Antimicrobial Susceptibility Testing (EUCAST) suggests the following epidemiological cut-off values (ECOFF mg/l) derived from MIC distribution curves to indicate susceptibility to topical moxifloxacin:

Corynebacterium: ND

Staphylococcus aureus: 0.25 mg/l

Staphylococcus, coag-neg.: 0.25 mg/l

Streptococcus pneumoniae: 0.5 mg/l

Streptococcus pyogenes: 0.5 mg/l

Streptococcus, viridans group: 0.5 mg/l

Enterobacter spp.: 0.25 mg/l

Haemophilus influenzae: 0.125 mg/l

Klebsiella spp.: 0.25 mg/l

Moraxella catarrhalis: 0.25 mg/l

Morganella morganii: 0.25 mg/l

Neisseria gonorrhoeae: 0.032 mg/l

Pseudomonas aeruginosa: 4 mg/l

Serratia marcescens: 1 mg/l

The prevalence of acquired resistance may vary geographically and with time for selected species and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of moxifloxacin in at least some types of infections is questionable.

COMMONLY SUSCEPTIBLE SPECIES

Aerobic Gram-positive micro-organisms:

Corynebacterium species including

Corynebacterium diphtheriae

Staphylococcus aureus (methicillin susceptible)

Streptococcus pneumoniae

Streptococcus pyogenes

Streptococcus viridans Group

Aerobic Gram-negative micro-organisms:

Enterobacter cloacae

Haemophilus influenzae

Klebsiella oxytoca

Moraxella catarrhalis

Serratia marcescens

Anaerobic micro-organisms:

Propionibacterium acnes

Other micro-organisms:

Chlamydia trachomatis

SPECIES FOR WHICH ACQUIRED RESISTANCE MAY BE A PROBLEM

Aerobic Gram-positive micro-organisms:

Staphylococcus aureus (methicillin resistant)

<i>Staphylococcus</i> , coagulase-negative species (methicillin resistant) Aerobic Gram-negative micro-organisms: <i>Neisseria gonorrhoeae</i> Other micro-organisms: None
INHERENTLY RESISTANT ORGANISMS
Aerobic Gram-negative micro-organisms: <i>Pseudomonas aeruginosa</i> Other micro-organisms: None

PHARMACOKINETICS:

Following topical ocular administration of Moxifloxacin, moxifloxacin was absorbed into the systemic circulation. The mean steady-state C_{max} and AUC were 2.7 ng/ml and 41.9 ng•hr/ml, respectively. These exposure values are approximately 1,600 and 1,200 times lower than the mean C_{max} and AUC reported after therapeutic 400 mg oral doses of moxifloxacin. The plasma half-life of moxifloxacin was estimated to be 13 hours.

INDICATIONS

Moxifloxacin Solution is indicated for the treatment of patients 1 year of age and older with bacterial conjunctivitis caused by susceptible strains of the following organisms.

Gram-positive bacteria:

Corynebacterium species†

Microbacterium species

Micrococcus luteus† [including erythromycin, gentamicin, tetracycline, and/or trimethoprim resistant strains]

Staphylococcus aureus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus epidermidis [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus haemolyticus [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains]

Staphylococcus hominis† [including methicillin, erythromycin, gentamicin, ofloxacin, tetracycline and/or trimethoprim resistant strains] *Staphylococcus warneri*† [including erythromycin resistant strains]

Streptococcus mitis† [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains] *Streptococcus pneumoniae* [including penicillin, erythromycin, gentamicin, tetracycline and/or trimethoprim resistant strains]

Streptococcus viridans [including penicillin, erythromycin, tetracycline and/or trimethoprim resistant strains]

Gram-negative bacteria:

Acinetobacter species

Haemophilus "alconae" [including ampicillin resistant strains]

Haemophilus influenzae [including ampicillin resistant strains]

Klebsiella pneumoniae†

Moraxella catarrhalis†

*Pseudomonas aeruginosa**

Other microorganisms:

Chlamydia trachomatis

†Efficacy for this organism was studied in fewer than 10 infections

Preoperative and postoperative sterilization (when prophylactic antibiotic treatment is required).

RECOMMENDED DOSAGE

Instill one drop in the affected eye 3 times a day for 7 days.

FOR PREOPERATIVE AND POSTOPERATIVE STERILIZATION: Usually, instill one drop in the affected eye(s) 5 times per day before operation, and 3 times per day after operation.

CONTRAINDICATIONS

Hypersensitivity to the active substance, to other quinolones, or to any of the excipients

WARNINGS AND PRECAUTIONS

In patients receiving systemically administered quinolones, serious and occasionally fatal hypersensitivity (anaphylactic) reactions have been reported, some following the first dose. Some reactions were accompanied by cardiovascular collapse, loss of consciousness, angioedema (including laryngeal, pharyngeal or facial oedema), airway obstruction, Dyspnoea, urticaria, and itching. If an allergic reaction to Moxifloxacin occurs, discontinue use of the medicinal product. Serious acute hypersensitivity reactions to moxifloxacin or any other product ingredient may require immediate emergency treatment. Oxygen and airway management should be administered where clinically indicated.

As with other anti-infective, prolonged use may result in overgrowth of non-susceptible organisms, including fungi. If super infection occurs, discontinue use and institute alternative therapy.

Tendon inflammation and rupture may occur with systemic fluoroquinolones therapy including moxifloxacin, particularly in older patients and those treated concurrently with corticosteroids. Following ophthalmic administration of Moxifloxacin plasma concentrations of moxifloxacin are much lower than after therapeutic oral doses of moxifloxacin, however, caution should be exercised and treatment with Moxifloxacin should be discontinued at the first sign of tendon inflammation.

Moxifloxacin should not be used for the prophylaxis or empiric treatment of gonococcal conjunctivitis, including gonococcal ophthalmic neonatorum, because of the prevalence of fluoroquinolones-resistant *Neisseria gonorrhoeae*. Patients with eye infections caused by *Neisseria gonorrhoeae* should receive appropriate systemic treatment.

Patients should be advised not to wear contact lenses if they have signs and symptoms of a bacterial ocular infection.

Paediatric population

Data are very limited to establish efficacy and safety of Moxifloxacin in the treatment of conjunctivitis in neonates. Therefore use of this medicinal product to treat conjunctivitis in neonates is not recommended.

Neonates with ophthalmia neonatorum should receive appropriate treatment for their condition, e.g. systemic treatment in cases caused by *Chlamydia trachomatis* or *Neisseria gonorrhoeae*.

The medicinal product is not recommended for the treatment of *Chlamydia trachomatis* in patients less than 2 years of age as it has not been evaluated in such patients. Patients older than 2 years of age with eye infections caused by *Chlamydia trachomatis* should receive appropriate systemic treatment.

Exacerbation of myasthenia gravis

Fluoroquinolones have neuromuscular blocking activity and may exacerbate muscle weakness in person with myasthenia gravis. Post marketing serious adverse events, including deaths and requirement for ventilator support have been associated with fluoroquinolones use in persons with myasthenia gravis. Avoid fluoroquinolones in patients with known history of myasthenia gravis.

INTERACTIONS WITH OTHER MEDICAMENTS

No specific interaction studies have been performed with Moxifloxacin 0.5%w/v eye drops, solution. Given the low systemic concentration of moxifloxacin following topical ocular administration of the medicinal product, drug interactions are unlikely to occur.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION

Pregnancy: There are no adequate data from the use of Moxifloxacin in pregnant women. However, no effects on pregnancy are anticipated since the systemic exposure to moxifloxacin is negligible. The medicinal product can be used during pregnancy.

Lactation: It is unknown whether moxifloxacin/metabolites are excreted in human milk. Animal studies have shown excretion of low levels in breast milk after oral administration of moxifloxacin. However, at therapeutic doses of Moxifloxacin no effects on the suckling child are anticipated. The medicinal product can be used during breast-feeding.

Fertility: Studies have not been performed to evaluate the effect of ocular administration of Moxifloxacin on fertility.

ADVERSE EFFECTS/ UNDESIRABLE EFFECTS

The following adverse reactions are classified according to the following convention: very common, common, uncommon, rare, very rare or not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in decreasing order of seriousness.

System Organ Classification	Frequency	Adverse reactions
Blood and lymphatic system disorders	Rare	haemoglobin decreased
Immune system disorders	Not known	hypersensitivity
Nervous system disorders	Uncommon	headache
	Rare	paresthesia
	Not known	dizziness
Eye disorders	Common	eye pain, eye irritation
	Uncommon	punctate keratitis, dry eye, conjunctival haemorrhage, ocular hyperaemia, eye pruritus, eyelid oedema, ocular discomfort,
	Rare	corneal epithelium defect, corneal disorder, conjunctivitis, blepharitis, eye swelling, conjunctival oedema, vision blurred, visual acuity reduced, asthenopia, erythema of eyelid
	Not known	endophthalmitis, ulcerative keratitis, corneal erosion, corneal abrasion, intraocular pressure increased, corneal opacity, corneal infiltrates, corneal deposits, eye allergy, keratitis, corneal oedema, photophobia, , eyelid oedema, lacrimation increased, eye discharge, foreign body sensation in eyes
Cardiac disorders	Not known	palpitations
Respiratory, thoracic and mediastinal disorders	Rare	nasal discomfort, Pharyngolaryngeal pain, sensation of foreign body (throat)
	Not known	Dyspnoea
Gastrointestinal disorders	Uncommon	dysgeusia
	Rare	vomiting
	Not known	nausea

Hepatobiliary disorders	Rare	alanine aminotransferase increased, gamma-glutamyltransferase increased
Skin and subcutaneous tissue disorders	Not known	erythema, rash, pruritus, urticaria

Description of selected adverse reactions

Serious and occasionally fatal hypersensitivity (anaphylactic) reactions, some following first dose, have been reported in patients receiving systemic quinolone therapy. Some reactions were accompanied by cardiovascular collapse, loss of consciousness, angioedema (including laryngeal, pharyngeal or facial oedema), airway obstruction, Dyspnoea, urticaria and itching. Ruptures of the shoulder, hand, Achilles, or other tendons that required surgical repair or resulted in prolonged disability have been reported in patients receiving systemic fluoroquinolones. Studies and post marketing experience with systemic quinolones indicate that a risk of these ruptures may be increased in patients receiving corticosteroids, especially geriatric patients and in tendons under high stress, including Achilles tendon.

Paediatric population:

Moxifloxacin has shown to be safe in paediatric patients, including neonates. The type and severity of adverse reactions in the paediatric population are similar to those in adults.

Exacerbation of myasthenia gravis

Post Marketing Experience

Effects on Ability to Drive and Use Machine

Moxifloxacin has no or negligible influence on the ability to drive and use machines, however, as with any eye drops, temporary blurred vision or other visual disturbances may affect the ability to drive or use machines. If blurred vision occurs at instillation, the patient should wait until their vision clears before driving or using machinery.

OVERDOSAGE AND TREATMENT

The limited holding capacity of the conjunctival sac for ophthalmic products practically precludes any overdosing of the medicinal product. The total amount of moxifloxacin in a single container is too small to induce adverse effects after accidental ingestion.

STORAGE CONDITIONS:

Store below 30°C before and after opening the bottle. Keep out from the reach of children.
Use the solution within 28 days after opening the bottle.

DOSAGE FORMS AND PACKAGING AVAILABLE

Ophthalmic Solution

5ml in 5ml three piece natural sterile bottle with natural sterile open nozzle and Sterile HDPE Tan TS cap packed in a carton.

NAME AND ADDRESS OF MANUFACTURER/ PRODUCT REGISTRATION HOLDER

MICRO LABS LIMITED,
Plot No. 113-116, Phase IV, KIADB Industrial Area,
Bommasandra Bangalore 560 099

PRODUCT REGISTRATION HOLDER:

Healol Pharmaceuticals Sdn Bhd
74-3, Jalan Wangsa Delima 6, KLSC
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Malaysia

DATE OF REVISION OF PI

January 2022

Manufactured By:



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